

5-Month Minimum-Sufficient Thesis Roadmap

This roadmap defines a conservative, execution-focused plan to complete the Master's thesis within five months. It assumes zero prior background in software quality metrics and strictly follows the scope agreed with the supervisors: a comparative study of software quality metrics for developer reputation systems, plus a minimal smart-contract proof of concept for one metric.

Month 1 – Foundations & Scope Lock

- Study basic concepts of software quality metrics and developer reputation.
- Read a small number of survey and foundational papers.
- Create a personal cheat sheet summarizing metric categories and intuition.
- Select a shortlist of 4–6 candidate metrics.
- Define comparison criteria: stability, interpretability, resistance to gaming, long-term suitability.

Month 2 – Dataset & Experimental Design

- Select 5–10 open-source projects (preferably same language, multiple versions).
- Freeze dataset and versions.
- Set up a simple metric-extraction pipeline using existing tools.
- Compute metrics and store results in reproducible tables.

Month 3 – Comparative Analysis

- Analyze metric stability across versions and projects.
- Compare metrics using the defined criteria.
- Produce plots, tables, and preliminary interpretations.

Month 4 – Metric Selection & Smart-Contract PoC

- Select one metric based on empirical results and implementation feasibility.
- Implement a minimal smart-contract proof of concept for the selected metric.
- Validate correctness and repeatability of the implementation.

Month 5 – Writing & Finalization

- Write full thesis draft (introduction, background, methodology, analysis, PoC, conclusions).

- Revise based on supervisor feedback.
- Finalize formatting and prepare for submission and defense.